Problem 6

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Problem 1 *Find the error in the following "solution", correct the error (or errors) and evaluate the integral.*

Find $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x)\sqrt{1-\cos^2(x)}\,dx$.

"Solution":

Since $\sin(x) = \sqrt{1-\cos^2(x)}$, *let* $u = \sin(x)$. *Then* $du = \cos(x)dx$;

when $x = -\dfrac{\pi}{2}$, $u = \sin\left(-\dfrac{\pi}{2}\right) = -1$; *when* $x = \dfrac{\pi}{2}$, $u = \sin\left(\dfrac{\pi}{2}\right) = 1$. *Therefore*

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x)\sqrt{1-\cos^2(x)}\,dx = \int_{-1}^{1} u\cdot du = 0,$$

due to symmetry (integrating an odd function over a symmetric interval, $[-1,1]$*).*

Explanation (Recall: $\sin(x) \le 0$ in the fourth quadrant!)**.** The error is that for $-\dfrac{\pi}{2} \le x \le 0$
$\sqrt{1-\cos^2(x)} = \sqrt{\sin^2(x)} = |\sin(x)| = -\sin(x)$,
not" $\sin(x)$", which makes the rest of the "solution" completely wrong.

In order to correct this error, we can split the integral into two integrals

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x)\sqrt{1-\cos^2(x)}\,dx = \int_{-\frac{\pi}{2}}^{0} \cos(x)\sqrt{1-\cos^2(x)}\,dx + \int_{0}^{\frac{\pi}{2}} \cos(x)\sqrt{1-\cos^2(x)}\,dx$$

and evaluate them separately, i.e., in the first integral we have that $\sqrt{1-\cos^2(x)} = -\sin(x)$ and in the second that $\sqrt{1-\cos^2(x)} = \sin(x)$.

Or, we can use symmetry to make the computation easier.

Namely, we "integrate" an even function over a symmetric interval $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$. Therefore

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x)\sqrt{1-\cos^2(x)}\,dx = 2\int_{0}^{\frac{\pi}{2}} \cos(x)\sqrt{1-\cos^2(x)}\,dx.$$

Now we use the same substitution as in the "solution" above.

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x)\sqrt{1-\cos^2(x)}\,dx = 2\int_{0}^{\frac{\pi}{2}} \cos(x)\sqrt{1-\cos^2(x)}\,dx = 2\int_{0}^{1} u\cdot du =$$

$$2\left[\frac{u^2}{2}\right]_0^1 = 2 \cdot \frac{1}{2} = 1$$